



PRODUCT NAME

Heavy-duty Powder Detergent Formula

Description

Laundry powder detergent for washing clothes

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-500 kg
Processing Time:	2-4 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Sodium Carbonate	Sodium Carbonate	25%	Water softener and alkalinity builder
2	Sodium Sulfate	Sodium Sulfate	20%	Filler and bulking agent
3	Sodium Dodecylbenzene Sulfonate	Sodium Dodecylbenzene Sulfonate	12%	Primary surfactant for stain removal
4	Sodium Lauryl Sulfate	Sodium Lauryl Sulfate	10%	Secondary surfactant for foaming
5	Zeolite A	Zeolite	8%	Water softener and ion-exchanger
6	Sodium Silicate	Sodium Silicate	7%	Alkalinity booster and corrosion inhibitor
7	Sodium Percarbonate	Sodium Percarbonate	6%	Oxygen bleach for stain removal
8	TAED (Tetraacetylene diamine)	Tetraacetylenediamine	3%	Bleach activator
9	Optical Brighteners	Fluorescent Brightener 260	0.5%	Enhances fabric brightness
10	Sodium Carboxymethylcellulose	Sodium Carboxymethylcellulose	1%	Anti-redeposition agent
11	Enzymes Blend	Protease/Lipase/Amylase	0.5%	Catalyzes breakdown of protein, fats, and starches
12	Perfume	Fragrance	0.5%	Provides a pleasant scent
13	Anti-caking Agent	Silica	1.5%	Prevents clumping and ensures free flow
14	pH Buffer	Sodium Bicarbonate	5%	pH adjuster to maintain pH 7
Total			100%	

Manufacturing Process

Phase 1: Manufacturing Phase

Phase 2: Manufacturing Phase

Phase 3: Manufacturing Phase

Phase 4: Manufacturing Phase

Phase 5: Manufacturing Phase

Phase 6: Manufacturing Phase

Phase 7: Manufacturing Phase

Phase 8: Manufacturing Phase

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply as needed according to product instructions